

Smart Route Optimization System

```
#include <bits/stdc++.h>

using namespace std;

#define INF INT_MAX

int main() {
    int V, E;
    cout << "Enter number of cities: ";
    cin >> V;

    cout << "Enter number of roads: ";
    cin >> E;

    vector<vector<pair<int, int>>> adj(V);

    cout << "Enter edges (u v weight):\n";
    for (int i = 0; i < E; i++) {
        int u, v, w;
        cin >> u >> v >> w;
        adj[u].push_back({v, w});
        adj[v].push_back({u, w});
    }

    int src, dest;
    cout << "Enter source city: ";
    cin >> src;
```

```
cout << "Enter destination city: ";
```

```
cin >> dest;
```

```
vector<int> dist(V, INF), parent(V, -1);
```

```
priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>> pq;
```

```
dist[src] = 0;
```

```
pq.push({0, src});
```

```
while (!pq.empty()) {
```

```
    int u = pq.top().second;
```

```
    pq.pop();
```

```
    for (auto it : adj[u]) {
```

```
        int v = it.first;
```

```
        int weight = it.second;
```

```
        if (dist[u] + weight < dist[v]) {
```

```
            dist[v] = dist[u] + weight;
```

```
            parent[v] = u;
```

```
            pq.push({dist[v], v});
```

```
        }
```

```
    }
```

```
}
```

```
if (dist[dest] == INF) {
```

```
        cout << "\nNo path exists between " << src << " and " << dest << endl;
        return 0;
    }

    cout << "\nShortest path from " << src << " to " << dest << ":\n";
    cout << "Distance: " << dist[dest] << "\nPath: ";

    vector<int> path;
    for (int v = dest; v != -1; v = parent[v]) {
        path.push_back(v);
    }

    reverse(path.begin(), path.end());

    for (int node : path) {
        cout << node << " ";
    }

    cout << endl;

    return 0;
}
```

Output 1:

```
C:\Users\saisa\OneDrive\Docu x + v - □ ×
Enter number of cities: 5
Enter number of roads: 4
Enter edges (u v weight):
0 1 4
0 2 2
1 3 5
2 3 1
Enter source city: 0
Enter destination city: 3

Shortest path from 0 to 3:
Distance: 3
Path: 0 2 3

-----
Process exited after 26.02 seconds with return value 0
Press any key to continue . . . |
```

Output 2:

```
C:\Users\saisa\OneDrive\Docu x + v - □ ×
Enter number of cities: 5
Enter number of roads: 5
Enter edges (u v weight):
0 1 4
0 2 2
1 3 5
2 3 1
3 4 6
Enter source city: 0
Enter destination city: 3

Shortest path from 0 to 3:
Distance: 3
Path: 0 2 3

-----
Process exited after 39.26 seconds with return value 0
```